

ON A VARIANCE ASSOCIATED WITH THE DISTRIBUTION OF PRIMES IN ARITHMETIC PROGRESSIONS

R. C. VAUGHAN

[Received 1 September 1999; revised 2 May 2000]

1. Introduction

Let Λ denote von Mangoldt's function. That is, when n is of the form p^k where p is a prime number and k is a natural number then $\Lambda(n) = \log p$ and otherwise $\Lambda(n) = 0$. Further let

$$\psi(x, q, a) = \sum_{\substack{n \leq x \\ n \equiv a \pmod{q}}} \Lambda(n) \quad (1.1)$$

and let $V(x, q)$ denote the variance

$$V(x, q) = \sum_{\substack{a=1 \\ (a, q)=1}}^q \left| \psi(x, q, a) - \frac{x}{\phi(q)} \right|^2 \quad (1.2)$$

and write $V_1(x, Q)$ for

$$V_1(x, Q) = \sum_{q \leq Q} V(x, q). \quad (1.3)$$

Suppose that

$$Q \leq x. \quad (1.4)$$

Then, as a refinement of work of Barban [1], Davenport and Halberstam [3], Gallagher [5] and Montgomery [12], Hooley [9, 10, 11] established the asymptotic formulae

$$V_1(x, Q) = Qx \log Q - cQx + O(Q^{5/4}x^{3/4} + x^2(\log x)^{-A}) \quad (Q \leq x) \quad (1.5)$$

and, on the assumption of the generalised Riemann hypothesis,

$$V_1(x, Q) = Qx \log Q - cQx + O(Q^{5/4}x^{3/4} + x^{3/2+\varepsilon}) \quad (Q \leq x). \quad (1.6)$$

Here

$$c = \gamma + \log(2\pi) + 1 + \sum_p \frac{\log p}{p(p-1)}. \quad (1.7)$$

Friedlander and Goldston [4] have obtained an improvement over (1.6), namely that on the generalised Riemann hypothesis

$$\begin{aligned} V_1(x, Q) &= Qx \log Q - cQx \\ &\quad + O(\min(Q^{3/2}x^{1/2}(\log x)^{3/2}, Qx) + x^{3/2}(\log x)^6) \quad (Q \leq x). \end{aligned} \quad (1.8)$$

This research was supported by NSF grant DMS-9970632.

2000 *Mathematics Subject Classification*: 11N13.

Proc. London Math. Soc. (3) 82 (2001) 533–553. © London Mathematical Society 2001.

Finally Goldston and Vaughan [6] have shown on the generalised Riemann hypothesis that if

$$D(x, Q) = V_1(x, Q) - Qx \log Q + cxQ, \quad (1.9)$$

then

$$D(x, Q) \ll Q^2(x/Q)^{1/4+\varepsilon} + x^{3/2}(\log 2x)^{5/2}(\log \log 3x)^2 \quad (1.10)$$

and that there is an absolute constant C such that when $x/Q \rightarrow \infty$ with

$$Cx^{5/7}(\log 2x)^{10/7}(\log \log 3x)^{8/7} < Q \leq x$$

one has

$$D(x, Q)Q^{-2} = \Omega_{\pm}((x/Q)^{1/4}). \quad (1.11)$$

They also indicate that without any unproved hypothesis their method will achieve

$$D(x, Q) \ll x^2(\log x)^{-A} + x^{1/2}Q^{3/2} \exp(-c'(\log(2x/Q))^{3/5}(\log \log(3x/Q))^{-1/5}). \quad (1.12)$$

In view of the above evidence, Hooley [8] has suggested that, for some unspecified range of q ,

$$V(x, q) \sim x \log q \quad (1.13)$$

and has shown, in [11], that if

$$x(\log x)^{-A} < Q \leq x \quad (1.14)$$

then (1.13) holds for almost all q in the range $\frac{1}{2}Q < q \leq Q$ and that on the generalised Riemann hypothesis the range (1.14) may be extended to

$$x^{4/5+\varepsilon} < Q \leq x. \quad (1.15)$$

Recently, Friedlander and Goldston [4] have shown that, on the generalised Riemann hypothesis, the range can be extended further, to

$$x^{3/4+\varepsilon} < Q \leq x. \quad (1.16)$$

More precisely they show, on the generalised Riemann hypothesis, that, when $x^{3/4} < Q \leq x$,

$$\sum_{Q/2 < q \leq Q} |V(x, q) - U(x, q)| \ll x^{7/4}(\log x)^{7/2} + \min\{Q^{3/2}x^{1/2}(\log x)^{3/2}, Qx\} \quad (1.17)$$

where

$$U(x, q) = x \log q - x \left(\gamma + \log 2\pi + \sum_{p|q} \frac{\log p}{p-1} \right). \quad (1.18)$$

They also indicate that, without the assumption of any unproved hypothesis, (1.17) can be inferred with the first term replaced by $x^2(\log x)^{-A}$ where A is an arbitrary fixed positive number.

In this memoir we develop the subject further by treating the general moment

$$M_k(x, Q) = \sum_{Q/2 < q < Q} |V(x, q) - U(x, q)|^k. \quad (1.19)$$

The ability to treat a general moment is a consequence of the observation that the underlying additive problem is tractable by a variant of the Hardy–Littlewood method. However, the conclusion also embraces a further refinement in the treatment of that part which dominates when Q is relatively close to x . This is intimately connected with the behaviour of the ‘partial singular series’

$$S_q(y) = \sum_{h \leq y} (y-h) \prod_{\substack{p|h \\ p \nmid 2q}} \frac{p-1}{p-2}. \quad (1.20)$$

There is a barrier on the line $\Re s = -\frac{1}{2}$ to the more obvious treatments (see, for example, the proof of Proposition 3 of Friedlander and Goldston [4]) and a substantial portion of this memoir is taken up with demolishing this barrier.

THEOREM 1. *Suppose that A is a positive number and that k is a positive integer. Then for every Q and x with $x(\log x)^{-A} \leq Q \leq x$ we have*

$$M_k(x, Q) = O_k(Qx^k F(x/Q)^k) + O_{k,A}(Qx^k (\log x)^{-A})$$

where, for $y \geq 1$,

$$F(y) \ll y^{-1/2} \exp\left(-c \frac{(\log 2y)^{3/5}}{(\log \log 3y)^{1/5}}\right)$$

with c a positive constant.

On the assumption of the Riemann hypothesis one can show that

$$F(y) \ll y^{\varepsilon-7/12}.$$

To make a concomitant improvement in the other term we require a more general hypothesis.

THEOREM 2. *Suppose that ε is a positive number, that k is an integer with $k \geq 2$ and that the Generalised Riemann Hypothesis holds. Then for every Q and x with $x^{3/4+\varepsilon} \leq Q \leq x$ we have*

$$M_k(x, Q) = O_k(Qx^k F(x/Q)^k) + O_{k,\varepsilon}(Qx^k (x/Q)^k x^{\varepsilon-1/2})$$

where, for $y \geq 1$,

$$F(y) \ll y^{\varepsilon-7/12}.$$

This gives a bound of the form $o(Qx^k)$ when $x/Q \rightarrow \infty$ and $Qx^{-1+1/(2k)-\varepsilon} \rightarrow \infty$ as $x \rightarrow \infty$. Both the terms in Theorem 2 are likely to be susceptible to some small improvements, although of most interest would be a bound $o(Qx^k)$ valid for values of Q in which the exponent $1 - 1/(2k)$ is replaced by something which remains substantially below 1 as k increases.

Since the new estimate for $S_q(y)$, defined in (1.20) above, is of some independent significance we display this as follows.

THEOREM 3. *Suppose that $y \geq 1$,*

$$C = \prod_{\substack{p > 2 \\ p \nmid y}} \frac{p(p-2)}{(p-1)^2}, \quad (1.21)$$

$$R_1 = C^{-1} \left(\prod_{\substack{p \mid y \\ p > 2}} \frac{p(p-2)}{(p-1)^2} \right) \frac{y^2}{2} \quad (1.22)$$

and

$$R_0 = -\frac{y}{4} C^{-1} \left(\prod_{\substack{p \mid y \\ p > 2}} \frac{p-2}{p-1} \right) \left(\log y + \gamma - 1 + \log(2\pi) + \sum_{p \mid 2y} \frac{\log p}{p-1} \right). \quad (1.23)$$

Then there is a positive constant c such that

$$S_q(y) = R_1 + R_0 + O \left(y^{1/2} \exp \left(\frac{-c(\log 2y)^{3/5}}{(\log \log 3y)^{1/5}} \right) \prod_{p \mid y} (1 - p^{-1/4})^{-1} \right). \quad (1.24)$$

We suppose throughout the rest of the memoir that $k \geq 2$, for the case $k = 1$ of Theorem 1 will follow from any higher moment by Hölder's inequality. Also, implicit constants may depend on k , c will denote a suitably small absolute constant, not necessarily the same one on different appearances, and ε will denote any sufficiently small positive number, not necessarily the same one on different occasions, and implicit constants in formulae in which ε appears may depend on ε .

We prove Theorem 1 in §§ 2–6 and then, in §§ 7–8 indicate the modifications necessary to prove Theorem 2.

2. Reduction to an additive problem

We insert the definition of $\psi(x, q, a)$ in (1.2) and square out. Thus

$$V(x, q) = W_0(x, q) - E_1(x, q) - \frac{x^2}{\phi(q)} - \frac{2x}{\phi(q)} (E(x) - E_2(x, q))$$

where

$$W_0(x, q) = \sum_{m \leq x} \sum_{\substack{n \leq x \\ n \equiv m \pmod{q}}} \Lambda(m) \Lambda(n), \quad E(x) = \psi(x) - x$$

and, for $q \leq x$,

$$E_1(x, q) = \sum_{\substack{m \leq x \\ (m, q) > 1}} \sum_{\substack{n \leq x \\ n \equiv m \pmod{q}}} \Lambda(m) \Lambda(n) \ll (\log x)^2 \log q$$

and

$$E_2(x, q) = \sum_{\substack{n \leq x \\ (n, q) > 1}} \Lambda(q) \ll (\log x)(\log q).$$

The diagonal terms in the double sum $W_0(x, q)$ contribute

$$\begin{aligned} \sum_{n \leq x} \Lambda(n)^2 &= \sum_{n \leq x} \Lambda(n) \log n + O(x^{1/2} \log x) \\ &= x(\log x - 1) + E(x) \log x - \int_1^x \frac{E(u)}{u} du + O(x^{1/2} \log x). \end{aligned}$$

Hence

$$V(x, q) = 2W(x, q) + x(\log x - 1) - \frac{x^2}{\phi(q)} + E_3(x, q) \quad (2.1)$$

where

$$W(x, q) = \sum_{m < x} \sum_{\substack{m < n \leq x \\ m \equiv n \pmod{q}}} \Lambda(m) \Lambda(n) \quad (2.2)$$

and

$$E_3(x, q) = E(x) \log x - \int_1^x \frac{E(u)}{u} du - \frac{2x}{\phi(q)} E(x) + O(x^{1/2} \log x), \quad (2.3)$$

and so, by (1.19),

$$M_k(x, Q) = \sum_{Q/2 < q \leq Q} |2W(x, q) - U^*(x, q) + E_3(x, q)|^k \quad (2.4)$$

where

$$U^*(x, q) = \frac{x^2}{\phi(q)} - x \left(\log \frac{x}{q} + \gamma - 1 + \log(2\pi) + \sum_{p|q} \frac{\log p}{p-1} \right). \quad (2.5)$$

Moreover, by Val'fiš [15],

$$E_3(x, q) \ll \frac{x^2}{\phi(q)} \exp(-c(\log x)^{3/5} (\log \log x)^{-1/5}) \quad (2.6)$$

and, on the Riemann Hypothesis (see, for example, § 8 of Davenport [2])

$$E_3(x, q) \ll x^{1/2} (\log x)^3 + \frac{x^{3/2}}{\phi(q)} (\log x)^2. \quad (2.7)$$

Let

$$f(\alpha) = \sum_{n \leq x} \Lambda(n) e(\alpha n) \quad (2.8)$$

and

$$H(\beta, q) = \sum_{\substack{h \leq x \\ q|h}} e(\beta h). \quad (2.9)$$

Then

$$W(x, q) = \int_0^1 |f(\alpha)|^2 H(\alpha, q) d\alpha. \quad (2.10)$$

Let

$$P = (\log x)^B \quad (2.11)$$

where B is at our disposal, but ultimately will be chosen suitably in terms of A and

k . We define a typical major arc $\mathfrak{M}(r, b)$ when $1 \leq b \leq r \leq P$ and $(b, r) = 1$ by

$$\mathfrak{M}(r, b) = \{\alpha: |\alpha r - b| \leq Px^{-1}\}. \quad (2.12)$$

Let $\mathfrak{M}$ denote their union

$$\mathfrak{M} = \bigcup_{\substack{1 \leq b \leq r \leq P \\ (b, r) = 1}} \mathfrak{M}(r, b) \quad (2.13)$$

and take the minor arcs $\mathfrak{m}$ to be the complement of $\mathfrak{M}$ with respect to

$$\mathfrak{U} = (x^{-1}P, 1 + x^{-1}P]. \quad (2.14)$$

For $\mathfrak{A} = \mathfrak{M}$ or $\mathfrak{m}$, let

$$I(\mathfrak{A}, q) = \int_{\mathfrak{A}} |f(\alpha)|^2 H(\alpha, q) d\alpha. \quad (2.15)$$

Then

$$W(x, q) = I(\mathfrak{M}, q) + I(\mathfrak{m}, q). \quad (2.16)$$

Hence, by (2.4),

$$M_k(x, Q) \ll \sum_{j=1}^2 M_k^{(j)}(x, Q) + x^{2k} Q^{1-k} \exp\left(-c \frac{(\log x)^{3/5}}{(\log \log x)^{1/5}}\right) \quad (2.17)$$

where

$$M_k^{(1)}(x, Q) = \sum_{Q/2 < q \leq Q} |2I(\mathfrak{M}, q) - U^*(x, q)|^k \quad (2.18)$$

and

$$M_k^{(2)}(x, Q) = \sum_{Q/2 < q \leq Q} |I(\mathfrak{m}, q)|^k. \quad (2.19)$$

3. The minor arcs

By Parseval's identity,

$$\int_{\mathfrak{m}} |f(\alpha)|^2 H(\alpha, q) d\alpha \leq \frac{x}{q} \int_0^1 |f(\alpha)|^2 d\alpha \ll x^2 Q^{-1} \log x. \quad (3.1)$$

Hence, by (2.19), for $k \geq 2$,

$$M_k^{(2)}(x, Q) \ll (x^2 Q^{-1} \log x)^{k-2} M_2^{(2)}(x, Q) \quad (3.2)$$

and, by (2.9) and squaring out and interchanging the order of summation and integration,

$$\begin{aligned} M_2^{(2)}(x, Q) &= \int_{\mathfrak{m}} \int_{\mathfrak{m}} |f(\alpha) f(\beta)|^2 \\ &\quad \times \sum_{h \leq 2x/Q} \sum_{j \leq 2x/Q} \sum_{Q/2 < q \leq \min(Q, x/h, x/j)} e((\alpha h - \beta j)q) d\alpha d\beta. \end{aligned}$$

The innermost sum here is

$$\ll \min(Q, \|\alpha h - \beta j\|^{-1}),$$

and

$$|f(\alpha)f(\beta)|^2 \leq |f(\alpha)|^4 + |f(\beta)|^4.$$

Thus

$$M_2^{(2)}(x, Q) \ll \int_{\mathfrak{m}} \sum_{h \leq 2x/Q} \sum_{j \leq 2x/Q} |f(\alpha)|^4 d\alpha \int_{\mathfrak{m}} \min(Q, \|\alpha h - \beta j\|) d\beta.$$

The inner integral here is at most

$$\int_0^1 \min(Q, \|\alpha h - \beta j\|^{-1}) d\beta \ll \log Q$$

and so

$$M_2^{(2)}(x, Q) \ll x^2 Q^{-2} (\log x) \int_{\mathfrak{m}} |f(\alpha)|^4 d\alpha. \quad (3.3)$$

By Theorem 3.1 of Vaughan [16] and (2.11),

$$\int_{\mathfrak{m}} |f(\alpha)|^4 d\alpha \ll x^3 (\log x)^{9-B}.$$

Therefore, provided that

$$B \geq A(k+2) + k + 8, \quad (3.4)$$

we have, by (3.2) and (3.3),

$$M_k^{(2)}(x, Q) \ll Q x^k (\log x)^{-A}. \quad (3.5)$$

4. The transformation of the major arcs

Our initial intent, as is usual with the analysis of major arcs, is to replace the generating function $f(\alpha)$ on $\mathfrak{M}(r, b)$ by a suitable smooth approximation, in this instance

$$\frac{\mu(r)}{\phi(r)} v\left(\alpha - \frac{b}{r}\right) \quad (4.1)$$

where

$$v(\beta) = \sum_{n \leq x} e(\beta n) \quad (4.2)$$

and then to extend the interval of integration to a complete unit interval. By Lemma 3.1 of Vaughan [16],

$$f(\alpha) = \frac{\mu(r)}{\phi(r)} v\left(\alpha - \frac{b}{r}\right) + O(x \exp(-c(\log x)^{1/2}))$$

whenever $1 \leq b \leq r \leq P$, $(b, r) = 1$ and $\alpha \in \mathfrak{M}(r, b)$. Thus

$$\sum_{r \leq P} \sum_{\substack{b=1 \\ (b,r)=1}}^r \int_{\mathfrak{M}(b,r)} \left| |f(\alpha)|^2 - \frac{\mu(r)^2}{\phi(r)^2} \left| v\left(\alpha - \frac{b}{r}\right) \right|^2 \right| d\alpha \ll P^2 x \exp(-c(\log x)^{1/2}).$$

Hence, by (2.15) and (2.18),

$$M_k^{(1)}(x, Q) \ll \sum_{Q/2 < q \leq Q} |2J(x, q) - U^*(x, q)|^k + x^{2k} Q^{1-k} P^{2k} \exp(-kc(\log x)^{1/2}) \quad (4.3)$$

where

$$J(x, q) = \sum_{r \leq P} \sum_{\substack{b=1 \\ (b, r)=1}}^r \frac{\mu(r)^2}{\phi(r)^2} \int_{\mathfrak{N}(r)} |v(\beta)|^2 H\left(\beta + \frac{b}{r}, q\right) d\beta \quad (4.4)$$

and

$$\mathfrak{N}(r) = \{\beta: |\beta| \leq Pr^{-1}x^{-1}\}. \quad (4.5)$$

By Hölder's formula for Ramanujan's sum we have

$$\sum_{\substack{b=1 \\ (b, r)=1}}^r e\left(\frac{b}{r}hq\right) = \sum_{d|(r, hq)} d\mu(r/d) \quad (4.6)$$

and so, by (2.9),

$$\sum_{\substack{b=1 \\ (b, r)=1}}^r H\left(\beta + \frac{b}{r}, q\right) \ll \sum_{h \leq x/q} \sum_{d|(r, hq)} d\mu(r/d)^2.$$

Let $\mathfrak{L}(r) = [-\frac{1}{2}, \frac{1}{2}] \setminus \mathfrak{N}(r)$. Then, by (4.5),

$$\int_{\mathfrak{L}(r)} |v(\beta)|^2 d\beta \ll xrP^{-1}$$

and so

$$\begin{aligned} \sum_{r \leq P} \sum_{\substack{b=1 \\ (b, r)=1}}^r \frac{\mu(r)^2}{\phi(r)^2} \int_{\mathfrak{L}(r)} |v(\beta)|^2 H\left(\beta + \frac{b}{r}, q\right) d\beta \\ \leq \sum_{m|q, m \leq P} \frac{m^2 \mu(m)^2}{\phi(m)^2} x^2 q^{-1} P^{-1} (\log x)^2. \end{aligned}$$

Therefore the $\mathfrak{N}(r)$ in (4.4) can be replaced by $[-\frac{1}{2}, \frac{1}{2}]$ with an error

$$\ll \sum_{m|q, m \leq P} \frac{m^2 \mu(m)^2}{\phi(m)^2} x^2 q^{-1} P^{-1} (\log x)^2.$$

Hence, by (4.3) and (4.4),

$$M_k^{(1)}(x, Q) \ll N_k^{(1)}(x, Q) + x^{2k} Q^{1-k} P^{2k} \exp(-kc(\log x)^{1/2}) + E$$

where

$$N_k^{(1)}(x, Q) = \sum_{Q/2 < q \leq Q} |2J_1(x, q) - U^*(x, q)|^k \quad (4.7)$$

with

$$J_1(x, q) = \sum_{r \leq P} \frac{\mu(r)^2}{\phi(r)^2} \sum_{\substack{b=1 \\ (b, r)=1}}^r \int_{-1/2}^{1/2} |v(\beta)|^2 H\left(\left(\beta + \frac{b}{r}\right)q\right) d\beta,$$

and

$$E = \sum_{q \leq Q} \left(\sum_{m|q} \frac{m^2 \mu(m)^2}{\phi(m)^2} \right)^k x^{2k} Q^{-k} P^{-k} (\log x)^{2k} \\ \ll x^{2k} Q^{1-k} (\log x)^{2^k - 1 + 2k} P^{-k}.$$

Hence if $B \geq B_0(k, A)$, we have, by (2.11),

$$M_k^{(1)}(x, Q) \ll N_k^{(1)}(x, Q) + Qx^k (\log x)^{-A}. \quad (4.8)$$

By (4.6),

$$J_1(x, q) = \sum_{r \leq P} \frac{\mu(r)^2}{\phi(r)^2} \sum_{h \leq x/q} \sum_{d|(r, hq)} d\mu(r/d)([x] - hq).$$

Here the $[x]$ can be replaced by x with an error

$$\ll \sum_{r \leq P} \frac{\mu(r)^2}{\phi(r)^2} \sum_{h \leq x/q} \sum_{d|(r, hq)} d\mu(r/d)^2 \ll xq^{-1} \log x.$$

Therefore,

$$N_k^{(1)}(x, Q) \ll N_k^{(2)}(x, Q) + Qx^k (Q^{-1} \log x)^k \quad (4.9)$$

where

$$N_k^{(2)}(x, Q) = \sum_{Q/2 < q \leq Q} |2J_2(x, q) - U^*(x, q)|^k$$

and

$$J_2(x, q) = \sum_{r \leq P} \frac{\mu(r)^2}{\phi(r)^2} \sum_{h \leq x/q} \sum_{d|(r, hq)} d\mu(r/d)(x - hq).$$

On interchanging the order of summation and replacing r by sd we see that

$$J_2(x, q) = \sum_{h \leq x/q} (x - hq) \sum_{\substack{d|hq \\ d \leq P}} \frac{d\mu(d)^2}{\phi(d)^2} \sum_{\substack{s \leq P/d \\ (s, d)=1}} \frac{\mu(s)}{\phi(s)^2}.$$

Since

$$\sum_{s > y} \frac{\mu(s)}{\phi(s)^2} \ll y^{-1},$$

it follows that the innermost sum can be completed to infinity with a total error in J_2 of

$$\ll \sum_{m|q, m \leq P} \frac{m^2 \mu(m)^2}{\phi(m)^2} x^2 q^{-1} P^{-1} \log x.$$

Hence, again if $B > B_0(k, A)$, then

$$N_k^{(2)}(x, Q) \ll N_k^{(3)}(x, Q) + Qx^k (\log x)^{-A} \quad (4.10)$$

where

$$N_k^{(3)}(x, Q) = \sum_{Q/2 < q \leq Q} |2J_3(x, q) - U^*(x, q)|^k$$

and

$$J_3(x, q) = \sum_{h \leq x/q} (x - hq) \sum_{\substack{d|qh \\ d \leq P}} \frac{d\mu(d)^2}{\phi(d)^2} \sum_{\substack{s=1 \\ (s,d)=1}}^{\infty} \frac{\mu(s)}{\phi(s)^2}.$$

The infinite series in J_3 satisfies

$$\sum_{\substack{s=1 \\ (s,d)=1}}^{\infty} \frac{\mu(s)}{\phi(s)^2} = \begin{cases} C \prod_{\substack{p|d \\ p>2}} \frac{(p-1)^2}{p(p-2)} & \text{for } d \text{ even,} \\ 0 & \text{for } d \text{ odd,} \end{cases}$$

where C is given by (1.21). Therefore

$$N_k^{(3)}(x, Q) = \sum_{Q/2 < q \leq Q} |4CJ_4(x, q) - U^*(x, q)|^k \quad (4.11)$$

where

$$J_4(x, q) = \sum_{\substack{h \leq x/q \\ 2|hq}} (x - hq) \sum_{\substack{d|hq, 2 \nmid d \\ d > P/2}} \frac{\mu(d)^2}{\prod_{p|d} (p-2)}.$$

In J_4 we may discard the condition $d \leq \frac{1}{2}P$ with an error

$$\begin{aligned} &\ll \sum_{h \leq x/q} x \sum_{\substack{d|hq, 2 \nmid d \\ d \leq P/2}} \frac{\mu(d)^2 d}{P \prod_{p|d} (p-2)} \\ &\ll x^2 q^{-1} P^{-1} \prod_{\substack{p|q \\ p>2}} \left(\frac{2p-2}{p-2} \right) \sum_{\substack{n \leq x/q \\ 2 \nmid n}} \frac{\mu(n)^2}{\prod_{p|n} (p-2)} \\ &\ll x^2 (\log x) q^{-1} P^{-1} \prod_{\substack{p|q \\ p>2}} \frac{2p-2}{p-2}. \end{aligned}$$

Thus

$$N_k^{(3)}(x, Q) \ll N_k^{(4)}(x, Q) + Qx^k (\log x)^{-A} \quad (4.12)$$

where

$$N_k^{(4)}(x, Q) = \sum_{Q/2 < q \leq Q} \left| 4CJ_5(x, q) \prod_{\substack{p|q \\ p>2}} \frac{p-1}{p-2} - U^*(x, q) \right|^k \quad (4.13)$$

with

$$J_5(x, q) = \sum_{\substack{h \leq x/q \\ 2|hq}} (x - hq) \prod_{\substack{p|h \\ p \nmid 2q}} \frac{p-1}{p-2}.$$

Hence, by (4.8), (4.9), (4.10) and (4.12),

$$M_k^{(1)}(x, Q) \ll N_k^{(4)}(x, Q) + Qx^k (\log x)^{-A}. \quad (4.14)$$

By (1.20),

$$J_5(x, q) = \begin{cases} qS_q(x/q) & \text{when } 2 \mid q, \\ 2qS_q(x/(2q)) & \text{when } 2 \nmid q. \end{cases} \quad (4.15)$$

5. Theorem 3; the analysis of $S_q(y)$

Let

$$D_q(s) = \sum_{h=1}^{\infty} \frac{1}{h^s} \prod_{\substack{p \mid h \\ p \nmid 2q}} \frac{p-1}{p-2}. \quad (5.1)$$

Then $D_q(s)$ converges absolutely for $\Re s > 1$ and satisfies

$$D_q(s) = \zeta(s) \sum_{\substack{d=1 \\ (d, 2q)=1}}^{\infty} \frac{\mu(d)^2}{d^s} \prod_{p \mid d} \frac{1}{p-2} \quad (5.2)$$

in that region. Moreover (5.2) affords the analytic continuation to the region $\Re s > 0$ and it is evident that there $D_q(s)$ is analytic at every point except $s = 1$ where, by (1.21), it has a simple pole with residue

$$\prod_{\substack{p \nmid 2q}} \left(1 + \frac{1}{p(p-2)}\right) = C^{-1} \prod_{\substack{p \mid q \\ p > 2}} \frac{p(p-2)}{(p-1)^2}. \quad (5.3)$$

We further have in the latter half plane

$$D_q(s) = \frac{\zeta(s)\zeta(s+1)}{\zeta(2s+2)} F_q(s) \quad (5.4)$$

where

$$F_q(s) = \left(\prod_{p \mid 2q} \frac{p^{s+1}}{p^{s+1}+1} \right) \prod_{p \nmid 2q} \frac{(p^s(p-2)+1)p}{(p-2)(p^{s+1}+1)}, \quad (5.5)$$

and, by §§ 3.10, 3.11 of Titchmarsh [14], this in turn gives the analytic continuation of $D_q(s)$ throughout the region $\Re s > -1$. Apart from the point $s = 1$ already discussed, there is a simple pole at $s = 0$ and otherwise $D_q(s)$, by Richert [13] and Theorem 3.11 of Titchmarsh [14], is analytic in a region of the kind

$$\sigma \geq \frac{1}{2} - \frac{c}{(\log \tau)^{2/3} (\log \log \tau)^{1/3}}$$

where $\sigma = \Re s$, $t = \Im s$ and $\tau = 3 + |t|$. We observe in passing that $F_q(s)$ has a simple zero at each point

$$-\frac{\log(p-2)}{\log p} + \frac{i\pi(2k+1)}{\log p}$$

with $k \in \mathbb{Z}$, and so the line $\sigma = -1$ is a natural boundary of D_q .

We are particularly concerned with the behaviour of

$$D_q(s) \frac{y^{s+1}}{s(s+1)}.$$

By (5.3), this function has a simple pole at $s = 1$ with residue R_1 given by (1.22) and a double pole at $s = 0$ with residue

$$R = y \frac{\zeta(0)F_q(0)}{\zeta(2)} \left(\log y + \gamma - 1 + \frac{\zeta'}{\zeta}(0) - 2 \frac{\zeta'}{\zeta}(2) + \frac{F'_q}{F_q}(0) \right). \quad (5.6)$$

We have

$$\begin{aligned} \frac{F_q(0)}{\zeta(2)} &= \left(\prod_{p|2q} \frac{p-1}{p} \right) \prod_{p \nmid 2q} \frac{(p-1)^2}{p(p-2)}, \\ -2 \frac{\zeta'}{\zeta}(2) + \frac{F'_q}{F_q}(0) &= \sum_{p|2q} \frac{\log p}{p-1}, \end{aligned}$$

and by § 2.4 of Titchmarsh [14],

$$\zeta(0) = -\frac{1}{2}, \quad \frac{\zeta'(0)}{\zeta(0)} = \log 2\pi.$$

Hence, by (1.21), (1.23) and (5.6), the residue at $s = 0$ is R_0 .

Let g be the multiplicative function defined by

$$g(p^k) = \begin{cases} \frac{(-1)^k}{p^k} & \text{when } p \mid 2q, \\ \frac{2(-1)^{k-1}}{p^k(p-2)} & \text{when } p \nmid 2q. \end{cases} \quad (5.7)$$

Then

$$F_q(s) = \sum_{n=1}^{\infty} \frac{g(n)}{n^s} \quad (5.8)$$

which converges absolutely for $\Re s > -1$. Moreover,

$$\sum_{n=1}^{\infty} \frac{|g(n)|}{n^{\sigma}} \ll \prod_{p|q} (1 - p^{-1/4})^{-1} \quad (5.9)$$

uniformly for $\sigma \geq -3/4$.

By the absolute convergence of $D_q(s)$ for $\sigma > 1$ we see that, by (1.20), when $y > 0$,

$$S_q(y) = \frac{1}{2\pi i} \int_{2-i\infty}^{2+i\infty} D_q(s) \frac{y^{s+1}}{s(s+1)} ds. \quad (5.10)$$

By § 5.1 of Titchmarsh [14],

$$\zeta(s) - \frac{1}{s-1} \ll \tau^{\lambda(\sigma)+\varepsilon} \quad (5.11)$$

where we take

$$\sigma = \Re s, \quad t = \Im s, \quad \tau = 3 + |t|, \quad \lambda(\sigma) = \begin{cases} 0 & \text{when } \sigma > 1, \\ \frac{1}{2} - \frac{1}{2}\sigma & \text{when } 0 < \sigma \leq 1, \\ \frac{1}{2} - \sigma & \text{when } \sigma \leq 0, \end{cases} \quad (5.12)$$

and, by § 3.11 of Titchmarsh [14],

$$\frac{1}{\zeta(s)} \ll \log \tau \quad \left(\sigma \geq 1 - \frac{c}{\log \tau} \right).$$

We henceforward suppose that $y \geq 1$ and let $T_0 = 8y^3$. Then the portions of the integral in (5.10) with $|\Im t| \geq T_0$ contribute an amount $\ll 1$. Moreover, by (5.12), when $\Im s = \pm T_0$,

$$D_q(s) \frac{y^{s+1}}{s(s+1)} \ll \prod_{p|q} (1 - p^{-1/4})^{-1}.$$

Therefore, by (5.10),

$$S_q(y) = R_1 + R_0 + \frac{1}{2\pi i} \int_{-1/2-iT_0}^{-1/2+iT_0} D_q(s) \frac{y^{s+1}}{s(s+1)} ds + O\left(\prod_{p|q} (1 - p^{-1/4})^{-1}\right). \quad (5.13)$$

In order to proceed further we need to replace T_0 by a much smaller quantity. We deal with the interval $\Im s \in [T_1, T_0]$ by first of all replacing each constituent factor of $D_q(s)$ by an approximating Dirichlet polynomial and then at the critical moment we apply the theory of exponential integrals.

Let T_1 be a parameter at our disposal, satisfying $0 < T_1 < T_0$. By (5.8) and absolute convergence, we have

$$\begin{aligned} \frac{1}{2\pi i} \int_{-1/2-iT_1}^{-1/2+iT_0} D_q(s) \frac{y^{s+1}}{s(s+1)} ds \\ = -\frac{1}{2\pi} \sum_{n=1}^{\infty} g(n)(yn)^{1/2} \int_{T_1}^{T_0} \frac{\zeta(-\frac{1}{2}+it)\zeta(\frac{1}{2}+it)(y/n)^{it}}{\zeta(1+2it)(\frac{1}{4}+t^2)} dt. \end{aligned} \quad (5.14)$$

By Richert [13] and Theorem 3.11 of Titchmarsh [14] we have

$$\frac{1}{\zeta(s)} \ll \log \tau \quad \left(\sigma \geq 1 - \frac{c}{(\log \tau)^{2/3}(\log \log \tau)^{1/3}}, \tau = 3 + |t| \right) \quad (5.15)$$

for a suitable small constant c . Thus, for a parameter $X \geq 2$ at our disposal, we have

$$\sum_{n \leq X} \frac{\mu(n)}{n^{1+iv}} (1 - n/X) = \frac{1}{2\pi i} \int_{1/\log X - i\infty}^{1/\log X + i\infty} \frac{X^w}{\zeta(1+iv+w)w(w+1)} dw.$$

Suppose that $W = 3 + 2|v|$. The portions of the integral with $|\Im w| > W$ contribute $\ll W^{-1} \log W$. We move the remaining interval to the vertical line with

$$\Re w = -\frac{c}{(\log W)^{2/3}(\log \log W)^{1/3}}.$$

In the process we pick up the residue $1/\zeta(1+iv)$ of the integrand at 1 and a further error, of amount $\ll W^{-2} \log W$ from the horizontal paths. Finally, the integral over the new vertical path is

$$\ll (\log W)^2 \exp \left(-\frac{c \log X}{(\log 2W)^{2/3}(\log \log 2W)^{1/3}} \right).$$

Now suppose that $T \geq 3$ and $T \leq t \leq 2T$. On choosing $v = 2t$ and

$$X = \exp(c'(\log T)^{5/3}(\log \log T)^{1/3})$$

for a suitable constant c' we see that

$$\frac{1}{\zeta(1+2it)} = \sum_{m \leq X} \frac{\mu(m)}{m^{1+2it}} (1 - m/X) + O(T^{-2/3}). \quad (5.16)$$

By (5.1.1) and Theorem 7.3 of Titchmarsh [14] we have, for $T \geq 3$,

$$\int_T^{2T} |\zeta(-\tfrac{1}{2}+it)\zeta(\tfrac{1}{2}+it)| dt \ll T^2 \log T.$$

Hence, by (5.16), for $3 \leq T \leq T' \leq 2T$,

$$\begin{aligned} & \int_T^{T'} \frac{\zeta(-\tfrac{1}{2}+it)\zeta(\tfrac{1}{2}+it)(y/n)^{it}}{\zeta(1+2it)(\tfrac{1}{4}+t^2)} dt \\ &= \sum_{m \leq X} \frac{\mu(m)}{m} \left(1 - \frac{m}{X}\right) \int_T^{T'} \frac{\zeta(-\tfrac{1}{2}+it)\zeta(\tfrac{1}{2}+it)(y/nm^2)^{it}}{\tfrac{1}{4}+t^2} dt + O(T^{-1/2}). \end{aligned} \quad (5.17)$$

By the functional equation for ζ (Titchmarsh [14, § 2.1]) we have

$$\begin{aligned} & \int_T^{T'} \frac{\zeta(-\tfrac{1}{2}+it)\zeta(\tfrac{1}{2}+it)(y/nm^2)^{it}}{\tfrac{1}{4}+t^2} dt \\ &= \sum_{l=1}^{\infty} l^{-3/2} \int_T^{T'} \frac{\chi(-\tfrac{1}{2}+it)\zeta(\tfrac{1}{2}+it)(yl/nm^2)^{it}}{\tfrac{1}{4}+t^2} dt \end{aligned} \quad (5.18)$$

where

$$\chi(s) = \pi^{s-1/2} \frac{\Gamma(\frac{1}{2}-\frac{1}{2}s)}{\Gamma(\frac{1}{2}s)}.$$

By Stirling's formula, for $t \geq 2$,

$$\chi(-\tfrac{1}{2}+it) = i(2\pi)^{-1} t \exp(-it \log t + it \log(2\pi) + it) + O(1).$$

Thus

$$\begin{aligned} & \int_T^{T'} \frac{\chi(-\tfrac{1}{2}+it)\zeta(\tfrac{1}{2}+it)(yl/nm^2)^{it}}{\tfrac{1}{4}+t^2} dt \\ &= \frac{i}{2\pi} \int_T^{T'} \frac{t \zeta(\tfrac{1}{2}+it) \exp(-it \log t + it \lambda)}{\tfrac{1}{4}+t^2} dt + O(T^{-1/2}), \end{aligned} \quad (5.19)$$

where

$$\lambda = \lambda(y, l, m, n) = 1 + \log \left(\frac{2\pi y l}{nm^2} \right).$$

By the approximate functional equation for ζ (Hardy and Littlewood [7]), when $t \geq 2$,

$$\zeta(\tfrac{1}{2}+it) = \sum_{j \leq \sqrt{t/2\pi}} j^{-1/2-it} + \chi(\tfrac{1}{2}+it) \sum_{j \leq \sqrt{t/2\pi}} j^{-1/2+it} + O(t^{-1/4}).$$

Therefore

$$\int_T^{T'} \frac{t \zeta(\frac{1}{2} + it) \exp(-it \log t + it \lambda)}{\frac{1}{4} + t^2} dt = I_1 + I_2 + O(T^{-1/4}) \quad (5.20)$$

where

$$I_1 = \sum_{j \leq \sqrt{T'/2\pi}} j^{-1/2} \int_{\max(T, 2\pi j^2)}^{T'} \frac{t \exp(-it \log t + it \lambda - it \log j)}{\frac{1}{4} + t^2} dt$$

and

$$I_2 = \sum_{j \leq \sqrt{T'/2\pi}} j^{-1/2} \int_{\max(T, 2\pi j^2)}^{T'} \frac{t \chi(\frac{1}{2} + it) \exp(-it \log t + it \lambda + it \log j)}{\frac{1}{4} + t^2} dt.$$

Again, by Stirling's formula,

$$\chi(\frac{1}{2} + it) = i \exp(-it \log t + it) + O(1/t)$$

and so

$$\begin{aligned} I_2 &= \sum_{j \leq \sqrt{T'/2\pi}} j^{-1/2} i \\ &\quad \times \int_{\max(T, 2\pi j^2)}^{T'} \frac{t \exp(it(-2 \log t + \lambda + 1 + \log j))}{\frac{1}{4} + t^2} dt + O(T^{-3/4}). \end{aligned}$$

By Lemma 4.5 of Titchmarsh [14], it follows that both I_1 and I_2 are bounded by

$$\sum_{j \leq \sqrt{T'/2\pi}} j^{-1/2} T^{-1/2} + T^{-3/4} \ll T^{-1/4}.$$

Therefore, by (5.14), (5.9), (5.20), (5.19), (5.18) and (5.17),

$$\frac{1}{2\pi i} \int_{-1/2+iT_1}^{-1/2+iT_0} D_q(s) \frac{y^{s+1}}{s(s+1)} ds \ll y^{1/2} T_1^{-1/4} (\log T_1)^2 \prod_{p|q} (1 - p^{-1/4})^{-1} \quad (5.21)$$

provided that $T_1 \geq 3$ and, of course, by taking complex conjugates we obtain the same bound for the contribution from the line segment joining $-\frac{1}{2} - iT_0$ to $-\frac{1}{2} - iT_1$. Therefore, by (5.13) and (5.21),

$$\begin{aligned} S_q(y) &= R_1 + R_0 + \frac{1}{2\pi i} \int_{-1/2-iT_1}^{-1/2+iT_1} D_q(s) \frac{y^{s+1}}{s(s+1)} ds \\ &\quad + O\left((1 + y^{1/2} T_1^{-1/4} (\log T_1)^2) \prod_{p|q} (1 - p^{-1/4})^{-1}\right). \end{aligned} \quad (5.22)$$

We now move the remaining vertical path into the critical region for $\zeta(2s+2)$. Let

$$\delta = c(\log(3 + 2T_1))^{-2/3} (\log \log(3 + 2T_1))^{-1/3} \quad (5.23)$$

where c is a suitable (small) positive constant. Then, by (5.11) and (5.12),

$$\zeta(s) \ll \tau^{1+\delta} \quad (\sigma \geq -\frac{1}{2} - \delta, \tau = 3 + |t|)$$

and

$$\zeta(s+1) \ll \tau^{1/3} \quad (\sigma \geq -\frac{1}{2} - \delta, \tau = 3 + |t|),$$

and by the functional equation for ζ , Stirling's formula and Theorem 7.2(A) of Titchmarsh [14],

$$\int_T^{2T} |\zeta(\frac{1}{2} - \delta + it)| dt \ll T^{1+\delta} (\log T)^{1/2}.$$

With these estimates and (5.15) it is readily seen on moving the vertical path to the line $\sigma = -\frac{1}{2} - \delta$ that

$$\begin{aligned} \frac{1}{2\pi i} \int_{-1/2-iT_1}^{-1/2+iT_1} D_q(s) \frac{y^{s+1}}{s(s+1)} ds \\ \ll (y^{1/2} T_1^{-1/4} + y^{1/2-\delta} T_1^{2\delta} (\log T_1)^3) \prod_{p|q} (1 - p^{-1/4})^{-1}. \end{aligned}$$

A good choice for T_1 (when $y > y_0$) is

$$T_1 = \exp \left(\frac{c'' (\log y)^{3/5}}{(\log \log y)^{1/5}} \right)$$

and Theorem 3 follows.

6. Completion of the proof of Theorem 1

By (4.15) and Theorem 3, when q is even,

$$\begin{aligned} 4CJ_5(x, q) \prod_{\substack{p|q \\ p>2}} \frac{p-1}{p-2} \\ = 4q \frac{x^2}{2q^2} \prod_{\substack{p|q \\ p>2}} \frac{p}{p-1} \\ - x \left(\log \frac{x}{q} + \gamma - 1 + \log(2\pi) + \sum_{p|q} \frac{\log p}{p-1} \right) + O(E_q(x/q)) \quad (6.1) \end{aligned}$$

where

$$E_q(y) = qy^{1/2} \exp \left(\frac{-c(\log y)^{3/5}}{(\log \log y)^{1/5}} \right) \prod_{\substack{p|q \\ p>2}} (1 - p^{-1/4})^{-1} \left(\frac{p-1}{p-2} \right). \quad (6.2)$$

Thus

$$\begin{aligned} 4CJ_5(x, q) \prod_{\substack{p|q \\ p>2}} \frac{p-1}{p-2} \\ = \frac{x^2}{\phi(q)} - x \left(\log \frac{x}{q} + \gamma - 1 + \log(2\pi) + \sum_{p|q} \frac{\log p}{p-1} \right) + O(E_q(x/q)). \end{aligned} \quad (6.3)$$

Likewise, when q is odd,

$$\begin{aligned} 4CJ_5(x, q) \prod_{\substack{p|q \\ p>2}} \frac{p-1}{p-2} \\ = \frac{x^2}{\phi(q)} - x \left(\log \frac{x}{2q} + \gamma - 1 + \log(2\pi) + \sum_{p|2q} \frac{\log p}{p-1} \right) + O(E_q(x/q)) \end{aligned}$$

where $E_q(x/q)$ again satisfies (6.2). Moreover the $\log 2$ arising from the term $\log(x/2q) = \log(x/q) - \log 2$ cancels with the term arising from the prime 2 in the sum. Thus (6.3) also holds when q is odd. Hence, by (4.13), (4.14), (2.5) and (1.18),

$$\begin{aligned} M_k^{(1)}(x, Q) &\ll \sum_{Q/2 < q \leq Q} E_q(x/q)^k + Qx^k(\log x)^{-A} \\ &\ll Q^{k+1}(x/Q)^{k/2} \exp\left(\frac{-kc(\log(2x/Q))^{3/5}}{(\log \log(3x/Q))^{1/5}}\right) + Qx^k(\log x)^{-A}. \end{aligned}$$

Hence, by (2.17) and (3.5), the theorem follows.

7. The assumption of the Generalised Riemann Hypothesis

Henceforward we assume the Generalised Riemann Hypothesis (GRH), that is, the hypothesis that no Dirichlet L -function $L(s, \chi) = \sum_{n=1}^{\infty} \chi(n)n^{-s}$ formed from a Dirichlet character modulo q has a zero ρ with $\Re \rho > \frac{1}{2}$.

The following lemma is routine, via the explicit formula for $\psi(x, \chi)$, partial summation and an application of standard exponential integral estimates.

LEMMA. Suppose that $(q, a) = 1$, $x \geq 1$, $|\beta| \ll (\log 2xq)^{-2}$ and $\alpha = aq^{-1} + \beta$. Then

$$f(\alpha) - \frac{\mu(q)}{\phi(q)} v\left(\alpha - \frac{a}{q}\right) \ll q^{1/2} x^{1/2} (\log 2qx)^2 + q^{1/2} x |\beta|^{1/2} (\log 2qx).$$

Since GRH will be our main tool regardless of the nature of the rational approximations to a given real number α , it is natural to not distinguish between major and minor arcs. We define a typical arc $\mathfrak{M}(r, b)$ as follows. Let

$$P = x^{1/2}(\log x)^{-1/2} \quad (7.1)$$

and consider the Farey fractions b/r of order P . Let $b_{\pm}/r_{\pm}$ denote the neighbouring fraction to the right and left respectively and put

$$\mathfrak{M}(r, b) = \left[\frac{b_- + b}{r_- + r}, \frac{b_+ + b}{r_+ + r} \right]. \quad (7.2)$$

Given b, r and P the values of $b_{\pm}, r_{\pm}$ are readily deduced from the observations $b_{\pm}r - r_{\pm}b = \pm 1$ and $P - r < r_{\pm} \leq P$. Thus

$$\mathfrak{M}(r, b) = \left[\frac{c_-}{s_-}, \frac{c_+}{s_+} \right] \quad (7.3)$$

where $-s_{\pm}b \equiv \pm 1 \pmod{r}$, $P < s_{\pm} \leq P+r$ and $c_{\pm} = (\pm 1 + s_{\pm}b)/r$, and so

$$\mathfrak{P}(r, b) \subset \mathfrak{M}(r, b) \quad (7.4)$$

where

$$\mathfrak{P}(r, b) = \left[\frac{b}{r} - \frac{1}{(2rP)}, \frac{b}{r} + \frac{1}{2rP} \right]. \quad (7.5)$$

Also

$$\mathfrak{M}(r, b) \subset \left[\frac{b}{r} - \frac{1}{(rP)}, \frac{b}{r} + \frac{1}{rP} \right]. \quad (7.6)$$

Let

$$\mathfrak{U} = \left[\frac{1}{1+[P]}, 1 + \frac{1}{1+[P]} \right]$$

and $\mathfrak{P}$ be the union of the $\mathfrak{P}(r, b)$ with $1 \leq b \leq r \leq P$ and $(r, b) = 1$. Then we define $V^*(\alpha)$ on $\mathfrak{U}$ by taking

$$V^*(\alpha) = \frac{\mu(r)}{\phi(r)} v(\alpha - b/r) \quad \text{when } \alpha \in \mathfrak{P}(r, b), \quad (7.7)$$

and define $V^{**}(\alpha)$ by taking

$$V^{**}(\alpha) = \begin{cases} V^*(\alpha) & \text{when } \alpha \in \mathfrak{P}, \\ 0 & \text{otherwise.} \end{cases} \quad (7.8)$$

Let

$$F_1(\alpha) = |f(\alpha) - V^*(\alpha)|^2, \\ F_2(\alpha) = 2\Re(f(\alpha)\overline{V^*(\alpha)} - |V^*(\alpha)|^2)$$

and

$$F_3(\alpha) = |V^*(\alpha)|^2 - |V^{**}(\alpha)|^2.$$

Then

$$|f(\alpha)|^2 = F_1(\alpha) + F_2(\alpha) + F_3(\alpha) + |V^{**}(\alpha)|^2.$$

By (2.4), (2.7) and the condition $x^{3/4+\varepsilon} \leq Q$, for $k \geq 2$,

$$M_k(x, Q) \ll \sum_{Q/2 < q \leq Q} |2W(x, q) - U^*(x, q)|^k + O(Qx^{k-1/2+\varepsilon}).$$

Therefore, by (2.10),

$$M_k(x, Q) \ll \sum_{j=1}^4 M_k^{(j)}(x, Q) + O(Qx^{k-1/2+\varepsilon}) \quad (7.9)$$

where

$$M_k^{(j)}(x, Q) = \sum_{Q/2 < q \leq Q} \left| \int_{\mathfrak{U}} F_j(\alpha) H(\alpha, q) d\alpha \right|^k \quad (1 \leq j \leq 3), \\ M_k^{(4)}(x, Q) = \sum_{Q/2 < q \leq Q} |2W_1(x, q) - U^*(x, q)|^k, \quad (7.10)$$

and

$$W_1(x, q) = \int_{\mathfrak{U}} |V^{**}(\alpha)|^2 H(\alpha, q) d\alpha. \quad (7.11)$$

It is a routine calculation to show that

$$\int_{\mathfrak{U}} |F_j(\alpha)| d\alpha \ll x \log x.$$

Thus for $j = 1, 2, 3$,

$$M_k^{(j)}(x, Q) \ll (x^2 Q^{-1} \log x)^{k-2} M_k^{(2)}(x, Q).$$

Moreover, by the Lemma, for $j = 1, 2$,

$$\int_{\mathfrak{U}} |F_j(\alpha)|^2 d\alpha \ll x^{5/2} (\log 2x)^4$$

and by a direct calculation an even stronger bound holds when $j = 3$. Clearly

$$M_2^{(j)}(x, Q) = \sum_{Q/2 < q \leq Q} \left| \sum_{\substack{h \leq x \\ q|h}} \int_{\mathfrak{U}} F_j(\alpha) e(h\alpha) d\alpha \right|^2.$$

By Cauchy's inequality this does not exceed

$$\sum_{Q/2 < q \leq Q} xq^{-1} \sum_{\substack{h \leq x \\ q|h}} \left| \int_{\mathfrak{U}} F_j(\alpha) e(h\alpha) d\alpha \right|^2 \ll x^{1+\varepsilon} Q^{-1} \sum_{h \leq x} \left| \int_{\mathfrak{U}} F_j(\alpha) e(h\alpha) d\alpha \right|^2$$

and by Bessel's inequality this does not exceed

$$x^{1+\varepsilon} Q^{-1} \int_{\mathfrak{U}} |F_j(\alpha)|^2 d\alpha.$$

Thus, for $j = 1, 2, 3$,

$$\begin{aligned} M_k^{(j)}(x, Q) &\ll (x^2 Q^{-1} \log x)^{k-2} x^{1+\varepsilon/2} Q^{-1} x^{5/2} (\log 2x)^4 \\ &\ll Q x^k (x/Q)^k x^{\varepsilon-1/2}. \end{aligned} \quad (7.12)$$

We now turn to $M_k^{(4)}(x, Q)$. The arguments of §§ 4, 5, with only modest adjustments, can be applied, starting from that which gives (4.8). Now we take $\mathfrak{N}(r) = \{\beta: |\beta| \leq r^{-1} P^{-1}\}$ and $\mathfrak{Q}(r) = [-\frac{1}{2}, \frac{1}{2}] \setminus \mathfrak{N}(r)$ and obtain, in place of (4.8),

$$M_k^{(4)} \ll N_k^{(1)}(x, Q) + Q(Q^{-1} x^{3/2+\varepsilon})^k \quad (7.13)$$

where $N_k^{(1)}(x, Q)$ satisfies (4.7). Then the argument that leads to (4.12) gives

$$N_k^{(1)}(x, Q) \ll N_k^{(4)}(x, Q) + Q(Q^{-1} x^{3/2+\varepsilon})^k \quad (7.14)$$

where $N_k^{(4)}(x, Q)$ satisfies (4.13).

In order to deal with $J_5(x, Q)$ we follow § 5 as far as (5.22). The proof of (5.16) could be simplified by an appeal to RH and a smaller value for X could be taken, but there seems to be no significant way to take advantage of RH until one reaches (5.22).

As usual we put $\tau = 3 + |t|$. Suppose that $-\frac{3}{4} + \varepsilon \leq \sigma \leq -\frac{1}{2}$. Then by § 5.1 of Titchmarsh [14] we have $\zeta(s) \ll \tau^{1/2-\sigma+\varepsilon}$, and by § 14.2 of Titchmarsh [14] we have

$$\zeta(s+1) \ll \tau^{-1/2-\sigma+\varepsilon} \quad \text{and} \quad \frac{1}{\zeta(2s+2)} \ll \tau^\varepsilon.$$

Now in the integral in (5.22) we replace the line segment $[-\frac{1}{2} - T_1, -\frac{1}{2} + T_1]$ by the line segments joining $-\frac{1}{2} - iT_1$, $-\frac{3}{4} + \varepsilon - iT_1$, $-\frac{3}{4} + \varepsilon + iT_1$, $-\frac{1}{2} + iT_1$ and apply the above estimates. Thus, by (5.4), (5.8) and (5.9),

$$\begin{aligned} \frac{1}{2\pi i} \int_{-1/2-iT_1}^{-1/2+iT_1} D_q(s) \frac{y^{s+1}}{s(s+1)} ds \\ \ll (y^{1/2} T_1^{3\varepsilon-1/2} + y^{1/4+\varepsilon} T_1^{1/2+3\varepsilon}) \prod_{p|q} (1 - p^{-1/4})^{-1}. \end{aligned}$$

Hence, by (5.22),

$$S_q(y) = R_1 + R_0 + O\left((1 + y^{1/2} T_1^{\varepsilon-1/4} + y^{1/4+\varepsilon} T_1^{1/2+\varepsilon}) \prod_{p|q} (1 - p^{-1/4})^{-1}\right).$$

A good choice for T_1 is $T_1 = 3y^{1/3}$ and this leads to

$$S_q(y) = R_1 + R_0 + O\left(y^{5/12+\varepsilon} \prod_{p|q} (1 - p^{-1/4})^{-1}\right).$$

It is readily seen that now we may follow §6 but with

$$E_q(y) = qy^{5/12+\varepsilon} \prod_{p|q} (1 - p^{-1/4})^{-1} \left(\frac{p-1}{p-2}\right).$$

Thus, by (7.9), (7.12), (7.13), (7.14) and (4.13),

$$M_k(x, Q) \ll Qx^k(x/Q)^k x^{\varepsilon-1/2} + Qx^k(x^{1/2+\varepsilon} Q^{-1})^k + Qx^k((x/Q)^{\varepsilon-7/12})^k$$

and Theorem 2 follows.

References

1. M. B. BARBAN, 'The large sieve method and its applications in the theory of numbers', *Uspekhi Mat. Nauk* 21 (1966) 51–102 (Russian), *Russian Math. Surveys* 22 (1966) 49–103 (English).
2. H. DAVENPORT, *Multiplicative number theory*, 2nd edn, Graduate Texts in Mathematics 74 (Springer, Berlin, 1980).
3. H. DAVENPORT and H. HALBERSTAM, 'Primes in arithmetic progressions', *Michigan Math. J.* 13 (1966) 485–489.
4. J. B. FRIEDLANDER and D. A. GOLDSTON, 'Variance of distribution of primes in residue classes', *Quart. J. Math. Oxford* (2) 47 (1996) 313–336.
5. P. X. GALLAGHER, 'The large sieve', *Mathematika* 14 (1967) 14–20.
6. D. A. GOLDSTON and R. C. VAUGHAN, 'On the Montgomery–Hooley asymptotic formula', *Sieve methods, exponential sums and their applications in number theory* (ed. G. R. H. Greaves, G. Harman and M. N. Huxley, Cambridge University Press, 1996) 117–142.
7. G. H. HARDY and J. E. LITTLEWOOD, 'The approximate functional equation for $\zeta(s)$ and $\zeta(s)^2$ ', *Proc. London Math. Soc.* (2) 29 (1929) 81–97.
8. C. HOOLEY, 'The distribution of sequences in arithmetic progression', *Proceedings of the International Congress of Mathematicians*, Vancouver, 1974, Vol. 1 (Canadian Mathematics Congress, Montreal, 1975) 357–364.
9. C. HOOLEY, 'On the Barban–Davenport–Halberstam theorem I', *J. Reine Angew. Math.* 274/275 (1975) 206–223.
10. C. HOOLEY, 'On the Barban–Davenport–Halberstam theorem II', *J. London Math. Soc.* (2) 9 (1975) 625–636.
11. C. HOOLEY, 'On the Barban–Davenport–Halberstam theorem III', *J. London Math. Soc.* (2) 11 (1975) 399–407.
12. H. L. MONTGOMERY, 'Primes in arithmetic progressions', *Michigan Math. J.* 17 (1970) 33–39.

13. H.-E. RICHERT, 'Zur Abschätzung der Riemannschen Zetafunktionen in der Nähe der Vertikalen $\sigma = 1$ ', *Math. Ann.* 169 (1967) 97–101.
14. E. C. TITCHMARSH, *The theory of the Riemann zeta-function*, 2nd edn revised by D. R. Heath-Brown (Clarendon Press, Oxford, 1986).
15. A. Z. VAL'FIŠ, *Weylsche Exponentialsummen in der neuen Zahlentheorie*, Mathematische Forschungsberichte XV (VEB Deutscher Verlag der Wissenschaften, Berlin, 1963).
16. R. C. VAUGHAN, *The Hardy–Littlewood method*, Cambridge Tracts in Mathematics 125 (Cambridge University Press, London, 1996).

R. C. Vaughan
Department of Mathematics
McAllister Building
Pennsylvania State University
University Park
PA 16802
USA
rvaughan@math.psu.edu